

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

29th October 2014

Subject CT6 – Statistical Methods

Time allowed: Three Hours (10.30 – 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1) i)** Individual claims from a block of general insurance policies are given by Y where $Y = X'$ & X follows $\text{LogNormal}(\mu, \sigma^2)$ distribution.
- Show that the probability that value of an individual claim lies between A & B ($B > A$) is $\exp(r\mu + r^2\sigma^2/2)[\phi(B_r) - \phi(A_r)]$, where $X_r = (\log X - \mu) / \sigma - r\sigma$, for $X = A, B$. $\phi(z)$ is the distribution function of the standard normal distribution. (6)
- ii)** Using the above results calculate the mean, variance & k th order non central moments of the Log Normal (μ, σ^2) distribution. (4)
- iii)** An insurer is considering taking out an unlimited individual excess of loss cover reinsurance treaty with an excess point of Rs 25,000. With the help of the above derived formula, find the mean & variance of the net claim amounts paid by the insurer, assuming that the distribution of individual losses is lognormal distribution with parameters $\mu = 8.5$ and $\sigma^2 = 0.64$. (8)
- [18]
- Q. 2)** A time series $\{X_t : t = 1, 2, \dots\}$ is defined as $X_t = \alpha X_{t-1} + Z_t$, for $|\alpha| < 1$ & Z_t are $N(0, \sigma^2)$ denoting white noise.
- i)** Express X_t as $X_t = \sum C_j Z_{t-j}$ where C_j 's are constant and the sum is taken for $j = 0$ to ∞ . (2)
- ii)** Hence, find expressions for the mean and variance of X_t in terms of α & σ^2 . (2)
- iii)** Explain why X_t has a normal distribution and hence calculate $P(X_{10} > 10)$ if $\alpha = 0.6$ and $\sigma^2 = 50$. (3)
- iv)** Without doing any further calculations, explain how the probability in part (iii) would change if $\alpha > 1$. (2)
- [9]
- Q. 3)** The number of claims, N , in a year on a portfolio of policies follow Poisson distribution with parameter λ . Large claims have a probability p and small claims $(1-p)$, independently of each other. Suppose we observe r large claims. Show that conditional distribution of $N-r|r$ is Poisson and find it's mean. (7)
- Q. 4)** In the country of Tyreia, a car tyre manufacturer offers a guarantee to purchasers of its tyres. There are 500 cars covered, each of which has a probability p of being involved in an accident (independently) and if a car is involved in an accident, there is a probability of 0.1 for each tyre (independently) that the tyre will need to be replaced at a cost of 5 units. Let S denote the total cost of replacement tyres in a year. (Assume each car has 4 tyres)
- i)** Find the moment generating function of S . (5)
- ii)** Hence show that $E(S) = 1000 p$ and $\text{Var}(S) = 6500 p - 2000p^2$ (6)
- Suppose instead that the manufacturer models the cost of replacement tyres as a random variable T based on a portfolio of 2000 tyres, each of which (independently) has a probability of $0.1p$ of requiring replacement.
- iii)** Derive expressions for $E(T)$ and $\text{Var}(T)$ in terms of p . (2)

iv) Suppose $p = 0.05$.

- a) Calculate the mean and variance of S and T .
- b) Calculate the probabilities that S and T exceed 75 units.
- c) Comment on the differences.

(5)

[18]

Q. 5) The number of phone calls received by a salesman each day has a Poisson distribution with mean λ , where from past experience, the prior distribution of λ is exponential with mean μ .

i) The company has data $x_1, \dots, x_n$, where x_i is the number of calls received on day i , $i = 1, \dots, n$.

- a) Derive the posterior distribution of λ .
- b) Show that the Bayesian estimate of λ under quadratic loss can be written in the form of a credibility estimate, and state the credibility factor.
- c) If $\mu = 50$ and the company receives a total of 650 calls over 10 days, calculate the Bayesian estimate of λ under quadratic loss.

(6)

ii) 60% of calls require an answering time (in minutes) that is exponentially distributed with mean 1, and the remaining messages require an answering time that has a Pareto distribution with mean 2 and variance 12.

- a) Determine the probability that a randomly chosen call requires more than M minutes answering time.
- b) Using the value of λ estimated in (i)(c), calculate the expected total amount of time the company spends answering all the calls that arrive on one day.
- c) The company decides to reduce the amount of time spent with calls and adopts a strategy of imposing a maximum cut-off time of 5 minutes for answering each call. Still using the estimated value of λ (assume that this restriction does not change the attitude of caller in respect to call time), calculate the reduction in the expected amount of time spent in answering the calls from one day.

(10)

[16]

Q. 6) A Marine insurer classifies the ships it insures into one of three types: Cargo, Passenger and Mixed. For Cargo ships, the number of claims per ship per year follows a Poisson distribution with parameter λ . Data is available for the last five years as follows:

Year	1	2	3	4	5
Number of Cargo ships insured	103	159	196	276	315
Number of claims	24	38	57	82	109

i) Determine the maximum likelihood estimate of λ based on the given data.

(5)

Based on the data collected by the insurer for all ships over the last five years. Define:

P_{ij} = the number of ships insured in j th year of type i and

Y_{ij} = the corresponding number of claims

The five years data is summarised as below:

Type (i)	$\bar{P}_i = \sum_{j=1}^5 P_{ij}$	$\bar{X}_i = \sum_{j=1}^5 \frac{Y_{ij}}{\bar{P}_i}$	$\sum_{j=1}^5 P_{ij} \left(\frac{Y_{ij}}{\bar{P}_i} - \bar{X}_i \right)^2$	$\sum_{j=1}^5 P_{ij} \left(\frac{Y_{ij}}{\bar{P}_i} - \bar{X} \right)^2$
Cargo	1049	0.295519542	1.719245636	2.710761957
Passenger	2282	0.245398773	2.420007402	3.276793898
Mixed	476	0.289915966	0.784583168	1.085437725

$$\bar{X} = \sum_{i=1}^3 \sum_{j=1}^5 \frac{Y_{ij}}{\bar{P}} = 0.264775 \text{ and } \bar{P} = \sum_{i=1}^3 \bar{P}_i = 3807$$

There are 400 cargo ships to be insured in year six.

- ii) Estimate the number of claims from Type 1 ship (cargo) in year six using Empirical Bayes Credibility Theory model 2. (6)
 - iii) Explain the main differences between the approaches in parts (i) and (ii). (3)
- [14]**

Q. 7) An insurer insures a single building. The number of claims are expected to follow a Poisson process with mean equal to μ . Premiums of 1 unit are payable on a daily basis at the start of each day. The claim size is independent of the time of the claim and follows an exponential distribution with mean $1/\lambda$. The insurer has a surplus of U at time zero.

- i) Derive an expression for the probability that the first claim results in the ruin of the insurer. (6)
 - ii) If $\mu = 0.01$ and $\lambda = 0.0125$ find how large U must be so that the probability that the first claim causes ruin is less than 1%. (2)
- [8]**

Q. 8) It is required to simulate observations from a random variable X with density function:

$$f(X) = 0.6 - 0.3x^2 \quad -1 < x < 1$$

using the acceptance-rejection method on a $U(-1,1)$ distribution.

- i) a) Use the pseudo-random numbers given below, which are uniform on $(0,1)$, to generate a set of three observations from the $U(-1,1)$ using the inverse transform method:
0.461, 0.966, 0.924
- b) Use the acceptance-rejection method on the observations from $U(-1,1)$ obtained in the part above to generate as many simulated values from X as possible. For the decision step, use the following additional pseudo-random numbers:
0.843, 0.427, 0.683 (8)
- ii) Determine the proportion of the simulations from the $U(-1,1)$ distribution that will be rejected in the long run. (1)
- iii) Given a reason why this method is preferable to a method based on inverting the cumulative distribution function of X . (1)

[10]
